

BANARAS HINDU UNIVERSITY
B.Sc. (Hons.) Semester III Examination, 2016-17

Zoology

Paper: ZOB-301 — Basic Genetics and Evolution & Economic Zoology

Time: Three hours

Full Marks: 70

Note: Write your Roll No. at the top immediately on the receipt of this question paper.

SECTION – A
(Basic Genetics and Evolution) (Marks: 35)

Note: Answer three questions including Question No. 1 which is compulsory.

Question 1. Answer the following parts:

(A) Select and write the correct choice in the following: [0.5 × 2 = 1]

- (i) A man of blood type A, whose mother had blood type O, marries a woman of blood type B, whose father was blood type O. The chances of this couple's having a child with blood type O are:

- (a) 25%
- (b) 50%
- (c) 75%
- (d) 100%

- (ii) A very dark skinned individual is mated with a light skinned individual. The F_1 individuals are all intermediate in skin colour. The F_2 individuals is variable; there are individuals as dark as the dark parent and as light as the light parent, with many other individuals showing many graduations in skin colour between the two extremes. What is the most likely mode of inheritance for skin colour?

- (a) pleiotropy
- (b) penetrance
- (c) epistasis
- (d) polygenic inheritance

(B) Write whether the following statements are true or false giving reasons for your answers in not more than 2–3 lines: [2 × 2 = 4]

- (i) Peking man belonged to *Homo habilis* species that originated in China.
- (ii) The phenotypic ratio of 1 : 2 : 1 is characteristic of the F_2 of a monohybrid cross where complete dominance is not there.

(C) Define in 2–3 lines: [1.5 × 2 = 3]

(i) Gloger's rule

(ii) Karyogram

(D) Fill in the blanks: [0.5 × 2 = 1]

(i) _____ is the condition of having an extra set of chromosomes in an organism.

(ii) When two heterozygous tall pea plants are crossed, _____ percent of the offsprings are short.

(E) Recombination frequency between two genes M and N is 16%. In a population of 200, how many recombinants and non-recombinants will be present? [2]

Question 2. Answer the following parts:

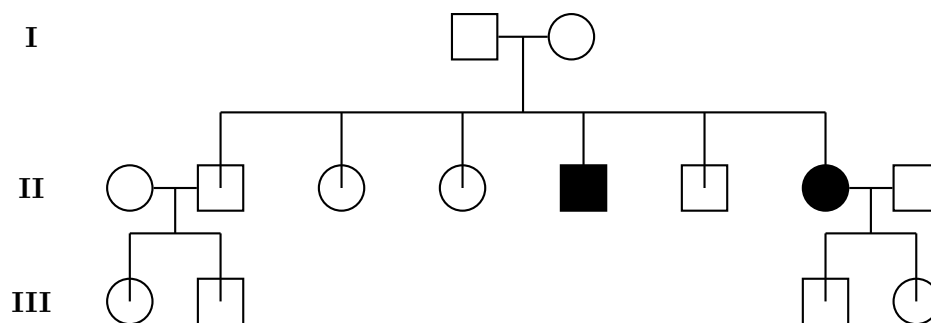
(A) Write short note on XX/XY sex chromosome system. [4]

(B) Briefly describe the phenomenon of cytoplasmic inheritance. [4]

(C) Very slight difference in survival brought about by minute changes in the genotype can result in substantial evolutionary change over a period of time. Explain. [4]

Question 3. Answer the following parts:

(A) A pedigree chart is shown below. Write whether the inheritance of the trait shown is sex linked or autosomal and dominant or recessive. What feature of the pedigree did you use to determine the inheritance. Write possible genotypes of each individual assuming the name of the alleles to be A and a: [6]



(B) Enlist the major structures observed in developing human embryo, which replicates its phylogenetic lineage. [4]

(C) Explain how lethal alleles disrupt Mendelian phenotypic ratios. [2]

Question 4. Answer the following parts:

(A) Illustrate meiotic consequences in an individual heterozygous for a reciprocal translocation. [6]

(B) Matings between purple starfish of identical genotype produced offspring as follows: 18 pink, 54 purple, and 24 albino.

(i) What epistatic ratio is shown by these offsprings? [3]

- (ii) What are the genotypes of the parents and the offspring (use your own symbols)?
[3]

SECTION – B
(Economic Zoology) (Marks: 35)

Note: Answer three questions including, Question No. 1 which is compulsory.

Question 1. Answer the following parts:

(A) Define the following: [1 × 2 = 2]

(i) Smart milk

(ii) Propolis

(B) State whether the following statements are True or False. Justify your answer in 2–3 sentences: [2 × 3 = 6]

(i) Worker honey bees are produced from the fertile eggs laid by the queen.

(ii) Pebrine is the disease of silk.

(iii) Lac is produced in large quantities by male lac insect.

(C) Answer the following: [1 × 3 = 3]

(i) Name any two breeds of indigenous cow.

(ii) Name any two species of exotic fish.

(iii) Name any two diseases of poultry.

Question 2. Answer the following parts:

(A) Describe various types of cattle sheds and their advantages. [6]

(B) Write a note on sericulture. [6]

Question 3. Answer the following parts:

(A) Give an account of induced breeding in fish. [6]

(B) Describe the different housing systems in poultry farming. [6]

Question 4. Write short notes on the following: [4 × 3 = 12]

(A) Lac processing and its uses

(B) Various castes of honey bee

(C) Pearl formation in natural condition.

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